

A Non-Empty Parameter Region for a Foliation-Based Relativistic MOND Theory: Eleven Gates Passed Below the Galaxy Scale, and a Factor 1.49–1.99 Mass Deficit Above It

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Abstract

A search over 666,082,872 grid points per acceleration footing finds, for the first time in this programme, a non-empty admissible region for a covariant modified-gravity action built on a preferred foliation: general relativity’s metric sector exactly, a khronometric clock scalar, and a MOND scalar sourced by the clock’s acceleration and carrying a coherence (healing-length) operator. An explicit point in that region — $K_B = 0.2$, $c_{14} = 1.0 \times 10^{-6}$, $\sigma = 1.6793$, $c_2 = 1.68 \times 10^{-6}$, $|K_2| = 1.93 \times 10^6$, $\xi = 0.10$ pc (canonical) / 0.15 pc (alt), $p = 1.7538$, and no condensate — passes all eleven admissibility gates on both of the framework’s acceleration footings ($a_0 = 9.3619 \times 10^{-11}$ and $1.1279 \times 10^{-10} \text{ m s}^{-2}$), and no subset of those gates up to size four is empty. The theory reproduces Milgrom’s law at the action’s own a_0 with coefficient exactly 1.000000, and with $\Phi = \Psi$ to better than 10^{-4} out to 1 Mpc, so lensing and dynamics share one potential with no slip and nothing fitted to make them do so. It passes the Solar-System gates (the Cassini quadrupole, the Pitjev–Pitjeva Saturn phantom-mass bound, the sunward-acceleration bound), the preferred-frame gates ($\alpha_1 = -4.5 \times 10^{-6}$, $\alpha_2 = -2.0 \times 10^{-7}$, $\alpha_3 = 0$, $\gamma = 1$), the tensor-speed gate ($c_{13} \equiv 0$, so $c_T = c$ exactly at every K_B), Hadamard well-posedness, gravitational Cherenkov, and causality — the clock’s leaves are spacelike for both cones, so no closed causal curve can be built even though the clock mode is 29.6% superluminal. It carries four propagating modes, all healthy, against the programme’s own requirement of two; that requirement, read as a total count, fails. **Above a galaxy the theory fails.** Its single fatal instability was a clock tachyon sourced by the scalar condensate’s background; curing it required setting the condensate amplitude to zero, and the condensate was the action’s only dark component. The theory is consequently short in mass at R_{500} by a factor 1.618 ± 0.022 (canonical) / 1.493 ± 0.020 (alt) against hydrostatic masses and 1.987 ± 0.246 / 1.834 ± 0.227 against weak lensing, the two probes agreeing with each other at 1.55σ , so no lensing-sector repair can reach the deficit; its $\Delta\Sigma$ log-slope is $+0.53 \pm 0.06$ shallower than measured, a 9σ error in the shape of the shear; the cluster-core acceleration excess of $3.37 a_0$ at 40 kpc exceeds the kernel’s own parameter-free ceiling by $5.2\times$; and with no dark component the CMB acoustic peaks, σ_8/S_8 and the matter power spectrum are unaddressed. Significances taken from samples that carry coherent systematics are labelled statistics-only and their systematics-floor values are given: the cluster shortfall’s $13\text{--}15\sigma$ becomes 4.4σ , the isolated binary-galaxy amplitude’s 19.6σ becomes 7.2σ or 3.0σ , and the Coma ultra-diffuse-galaxy amplitude (a factor of 14) is 4.9σ , not the 19.4σ once quoted; in each case the direction survives and only the number falls. The relation $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ is fitted, not derived [24]: $\kappa = 0.4998$ canonical and 0.6023 alt, so the tie does not even select a footing. Distinctive predictions are stated with numbers and falsifiers, among them a parameter-free ceiling on the acceleration excess and a deep-MOND baryonic Tully–Fisher zero point flat to better than 1% out to $z = 5$. Every number is produced by a committed

script with checks that can fail (`fable_independent_2026/L43_assemble_theory.py` and its supporting lanes in the public repository). AI-assisted research draft; not peer reviewed.

1 The headline, first

The parameter space is non-empty. After eighteen months in which no single point satisfied every constraint at once, there is now a region, and an explicit point inside it that passes eleven independent gates on both of the framework’s acceleration footings. The point is exhibited in §6 and every gate is tabulated in §7.

What that region buys is a complete relativistic theory of gravity for the Solar System and for galaxies, and nothing above a galaxy. Cassini, the Saturn phantom-mass bound, the sunward-acceleration bound, α_1 , α_2 , α_3 , γ , the tensor speed, the mode count and its health, Hadamard well-posedness, causality, black holes, rotation curves and the lensing/dynamics slip — all pass, at one point, on both footings. Clusters, ultra-diffuse galaxies, binary galaxies, the CMB and S_8 all fail, and they fail for one reason: the price of curing the theory’s single fatal instability was deleting its only dark component.

Stated the way it should be quoted:

A complete theory of gravity that works everywhere except above a galaxy, where it is short by a factor of 1.49–1.99 in mass at R_{500} — in lensing and in dynamics alike — with a 9σ error in the shape of $\Delta\Sigma$, and where its own hard acceleration ceiling is exceeded by $5.2\times$ at 40 kpc in cluster cores.

Nothing here is closed. $\kappa = \frac{1}{2}$ is **fitted**, not derived, and this paper never says otherwise.

Significances in this paper. Every σ quoted below is labelled where it is a statistics-only figure on a sample carrying coherent systematics. An adversarial audit of the underlying record (`L36_record_audit.py`) re-ran all 22 source scripts — every one reproduces its committed output to the character — and found 22 defects, **all of them errors of description rather than of computation**, of which three were significances computed as though correlated errors were independent. Those three are corrected inline here. In every case the direction survives and only the number falls.

2 The structural decision, and the theorem behind it

For a long time the extra propagating scalar in this class of theories [4, 3] was treated as a defect to be engineered away. Two results say it cannot be, and the theory is therefore built *on* a preferred foliation with a healthy clock mode, stated as a consequence rather than apologised for.

Theorem 1 (Lorentz invariance XOR two modes) *Under four hypotheses — (i) the static weak-field limit is Milgrom’s equation [1, 2], (ii) one metric, minimally coupled to matter, (iii) exactly two propagating gravitational modes, (iv) locality — a distinguished timelike vector field u is forced; adding the elliptic boundary-value posing of Milgrom’s equation, Frobenius makes it hypersurface-orthogonal, i.e. a foliation.*

Verified against a seventeen-theory control table (GR, Brans–Dicke, $f(R)$, TeVeS, Einstein-aether [5], khronometric [6], Hořava, AeST, ...): **no row has both Lorentz invariance and exactly two modes**, and no counterexample was found. Controls return the right mode counts 2/3/3/5 for GR, GR + scalar, khronometric and Einstein-aether. 53/53 checks pass. `L31_foliation_nogo.py`.

Companion theorem. No foliation-independent scalar can replace the slice-built field, on four independent counts. Fourteen candidates were tested on a *three-member* slicing family of vacuum Schwarzschild (static / tilted / Painlevé–Gullstrand) — a two-slicing test would have passed two of the losers by coincidence. Seven candidates die on slice-dependence alone. The decisive count (T3) is field-independent: on a homogeneous slice every spatial scalar is constant, so $\mu(0) = 0$ forces $\rho = 0$ against the measured $2.688 \times 10^{-27} \text{ kg m}^{-3}$. That kills the *structure*, not a choice within it. `L27_foliation_scalar.py`.

The one escape, named and left open. Hypothesis (iv), locality, is not proved. Temporal nonlocality is a genuine escape: holding the external field fixed and pushing the source $1000\times$ further away drops every local curvature invariant by $1000\times$ while $|\nabla(\Box^{-1}R)|$ is exactly constant. Deciding it requires a definitive Hamiltonian mode count for retarded-nonlocal gravity, which is being done elsewhere and is **not** claimed here. **If that escape opens, the foliation below is a choice, not a theorem.** Everything else in this paper is unaffected.

Consequence: the theory has **four** propagating modes — two tensor, one clock, one MOND scalar — not two. All four are healthy. That is recorded as a cost against the programme’s own “ $N_{\text{grav}} = 2$ ” requirement in §7, not hidden.

3 The action

Fields: a metric $g_{\mu\nu}$; a clock scalar τ , with unit normal $n_\mu = -\partial_\mu\tau/\sqrt{-g^{\alpha\beta}\partial_\alpha\tau\partial_\beta\tau}$ and spatial projector $q_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$; a MOND scalar ϕ , with $Q = n^\mu\partial_\mu\phi$, $V_\mu = q_\mu^\nu\partial_\nu\phi$, $Y = V\cdot V$. Matter is **minimally coupled to** $g_{\mu\nu}$ — one metric, no disformal or conformal factor on the matter sector.

$$\begin{aligned}
S = \int d^4x \sqrt{-g} \Big\{ & \frac{1}{16\pi G} [R - 2\Lambda] \\
& - c_1(\nabla_\mu n_\nu)(\nabla^\mu n^\nu) - c_2(\nabla\cdot n)^2 - c_3(\nabla_\mu n_\nu)(\nabla^\nu n^\mu) + c_4(n\cdot\nabla n)^2 \\
& + 2(2 - K_B) J^\mu \partial_\mu \phi - K(Q) \\
& - (2 - K_B) J \Big(Y + \xi^2 q^{\lambda\sigma} q^{\mu\nu} \nabla_\lambda V_\mu \nabla_\sigma V_\nu \Big) \Big\} + S_m[g, \psi_m], \tag{1}
\end{aligned}$$

with $J^\mu \equiv n^\nu \nabla_\nu n^\mu$. Line by line:

term	what it is	what it does
$\frac{1}{16\pi G}(R - 2\Lambda)$	Einstein–Hilbert with a cosmological constant	the metric sector is exactly Einstein on the operative branch (DeWitt $\lambda = 1$ identically, first-class Hamiltonian constraint)
$c_1 \dots c_4$	the aether/khronometric kinetic sector, with $c_1 = -c_3 = K_B$ so $c_{13} = 0$ identically	$c_{13} = 0$ makes $c_T = c$ <i>exactly</i> , at every K_B — GW170817 [7] is satisfied structurally, not by tuning
$2(2 - K_B)J^\mu \partial_\mu \phi$	the AeST coupling of the scalar gradient to the clock’s 4-acceleration	this is what sources the MOND scalar with matter : statically $\nabla \cdot J = \nabla^2 \Psi$
$K(Q)$	the clock-direction kinetic function of the MOND scalar, $K(Q) = K_2 Q^2$	sets the scalar’s time-kinetic stiffness. $Q_0 = 0$: there is no condensate (§4.2)
$J(Y + \xi^2 \nabla_\perp V ^2)$	the MOND function of the <i>spatial</i> gradient, plus the coherence (healing-length) operator [25]	J gives Milgrom’s law; ξ screens the Solar System. Both are forced, neither is chosen (§5)

With $c_{13} = 0$ and hypersurface-orthogonal n , the clock sector *is* khronometric theory with $\alpha = c_{14}$, $\lambda = c_2$, $\beta = 0$.

The static weak-field limit is Milgrom’s equation at the action’s own a_0 , with coefficient exactly 1.000000 and no slip. Independently re-derived: the clock’s auxiliary equation $d/dc[(1 - c)a^2 - a_0^2 U(c)] = 0$ with $U'(c) = -\ln^2(1 - c)$ solves to $u^2 = 1 - e^{-|a|/a_0}$ *exactly* on the static branch (not as a weak-field approximation), giving

$$\nabla \cdot [\mu(|\nabla \Phi|/a_0) \nabla \Phi] = 4\pi G_N \rho_b, \quad \Phi = \Psi, \quad G_{\text{dyn}} = G_{\text{lens}} = G_{\text{tensor}}. \quad (2)$$

$\Phi = \Psi$ to better than 10^{-4} out to 1 Mpc, so lensing and dynamics share one potential — the framework’s lensing requirement, met structurally rather than fitted. `L11_galactic_limit.py`; re-derived from scratch in `L43_assemble_theory.py` checks K4/K4b.

4 The parameters, with the constraint that fixes each

Both footings are carried on every dimensional number: $a_0 = 9.3619 \times 10^{-11} \text{ m s}^{-2}$ (**canonical**) and $1.1279 \times 10^{-10} \text{ m s}^{-2}$ (**alt**).

parameter	admissible window	what fixes it	script
$K_B (= c_1 = -c_3)$	$0 < K_B \leq 0.25$	BBN on the cosmological Newton constant	<code>route2, g03e B1</code>
$c_{13} = c_1 + c_3$	0 exactly	$c_T = c$ (GW170817). Identically zero by $K_B - K_B$	<code>structural</code>
$c_{14} = c_1 + c_4$	$\leq 1.98 \times 10^{-6}$	see the tightening below	<code>L13, L26, this script</code>
c_2 (khronometric λ)	fixed by σ : $c_2 = 2\sigma c_{14}/(2 - c_{14} - 3\sigma c_{14}) \approx \sigma c_{14}$	the clock's own speed	<code>L26, this script</code>
$\sigma = \text{clock speed}^2$	$[1.6793, 1.7716)$, with $\sigma_* = 1.679312732$	see below	<code>L26_sigma_</code>
$ K_2 $	on the closure locus $c_2 K_2 = (2 - K_B)^2$	linear growth $\wedge$ Cherenkov	<code>above_one.py</code>
ξ (coherence length)	$\geq 0.10 \text{ pc}$ canonical / 0.15 pc alt — theorem-forced	§5	<code>g03t D5/D7, g03v</code>
p (saturation exponent)	$0 < p \leq 1.754$	perturbativity of the cubic action at a planet	<code>L34, g03d, g03z</code>
Q_0 (condensate rate)	0 — the condensate is removed	see below	<code>L34_boost_vs_cubic.py</code>
a_0	$9.3619 \times 10^{-11} / 1.1279 \times 10^{-10} \text{ m s}^{-2}$	measurement, two footings	<code>L14, this script</code>
κ in $a_0 = \kappa c \sqrt{G\rho_\Lambda}$	0.4998 canonical / 0.6023 alt — FITTED	nothing. See §8.3	<code>—</code>
			<code>L32_kappa_necessary.py</code>

4.1 σ , and why it is above 1

The construction that generates the clock sector carries a design parameter σ — the clock mode's squared speed — declared over $(0, 1]$ in its own documentation. That declaration turns out to be an **undefended convention**: it appears once in prose, is copied ten more times, and no `assert`, `if`, `while` or `raise` anywhere in the construction mentions σ . The only executable restriction is $\sigma > 0$.

There is exactly one value of σ at which the construction's quartic obstruction — which otherwise produces a Hadamard-ill-posed growth rate rising like k^2 — has its leading order vanish:

$$\sigma_* = \frac{4T}{4T - 27} = \frac{3}{a_*}, \quad T = -\frac{27}{16} + \frac{54}{5 \ln(9/5)} = 16.6865133026, \quad a_* = 3 - \frac{81}{4T}, \quad (3)$$

$$\sigma_* = 1.679312732187113 \quad (\text{a clock } 29.6\% \text{ superluminal}). \quad (4)$$

At σ_* the obstruction's leading order is zero *and the sign of the residual reverses on all 59 tested branch points*, so the real growth rate becomes a bounded oscillation: **the ill-posedness is removed, not reduced**. All fourteen of the construction's own health conditions hold there — no ghost (the coefficient is exactly σ -free), positive mode energy, $A_0 = 0.4615 > 0$, $c_T^2 = 1$ as a σ -identity, hyperbolicity, an unmoved constraint fold, an unchanged $\det M_*$. The **only** internal ceiling anywhere in the family is $J_T > 0$, which is exactly affine in σ and vanishes at $\sigma = 1.7716$; σ_* sits 5.49% below it. So the admissible set is a *window*, $[1.6793, 1.7716)$, not a tuned point.

Superluminality is not a problem here, and this is worth stating clearly, because it is usually assumed to be one:

- **Causality.** The clock mode's acoustic metric is $G^{\mu\nu} = g^{\mu\nu} + (1 - 1/\sigma)n^\mu n^\nu$, so $G^{\mu\nu}n_\mu n_\nu = -1/\sigma < 0$ for every $\sigma > 0$. The clock's own leaves are spacelike for *both* cones, so τ is a global

time function for both and no closed causal curve can be built. This is available to *this* theory specifically, because the acoustic metric is built from the same foliation the theory already varies.

- **Gravitational Cherenkov.** The 2×10^{-15} bound [27, 9] is a **lower** bound: it constrains modes *slower* than an ultra-high-energy cosmic ray, because that is what a ray can radiate into. A 29.6% superluminal mode switches the channel off. (Independently, the bound’s applicability to a conformally coupled k -essence clock loosens by $7.2 \times 10^5 \times$ even for subluminal modes, and vanishes entirely once the high-acceleration limit [28] is applied in the cosmic ray’s own near field.)
- **Black holes.** The universal horizon at $r = 3M/2$ is reproduced; even an infinite-speed mode is trapped.

A new consequence, derived in the assembly. Tying c_2 to c_{14} through σ makes the preferred-frame PPN parameter

$$\alpha_2 \simeq \frac{c_{14}}{2} \left(\frac{1}{\sigma} - 1 \right) = -0.2023 c_{14} \quad \text{at } \sigma = \sigma_*, \quad (5)$$

so the Solar-System bound $|\alpha_2| < 4 \times 10^{-7}$ [8] forces $c_{14} \leq 1.978 \times 10^{-6}$ — twelve times tighter than the bound $\alpha_1 = -4c_{14}$ imposes on its own (2.5×10^{-5}), and six times tighter than a previous global grid scan found (1.18×10^{-5}). The three bounds are **nested, not in conflict**: three lanes, different machinery, same corner.

Caveat, stated: the identification $\sigma = (2 - c_{14})c_2/[c_{14}(2 + 3c_2)]$ between the construction’s design parameter and the khronometric sound speed is **stated, not proved** — the mapping from the construction’s variables to $(c_{14}, c_2, |K_2|)$ is not determined by its published files. The non-emptiness of the parameter region does **not** depend on it (§6); only the *sharpness* of the c_{14} bound does.

4.2 $Q_0 = 0$: the condensate is removed, and why that is the right move

A previous global sweep over 1.2×10^8 grid points per footing, with ten gates each reproducing its own source script’s published verdict as a control, found **no admissible point at all**. The obstruction was exactly two gates deep, and *every* minimal incompatible subset contained the same gate: the clock tachyon. The condensate’s background contributes a positive gradient term for the clock’s time shift, giving a k -independent instability with rate² $= |K_2|Q_0^2\varepsilon_0 a^{-3}/c_{14}$ — about $280 H_0$ today, $2.8 \times 10^5 H_0$ at $z = 100$. Stability needs $c_{14} \geq 3\Omega_d/\Omega_m = 2.533$; PPN needs $c_{14} \leq 2.5 \times 10^{-5}$. A $10^5 \times$ gap.

The escape is not parametric, it is structural, and it is free: **the condensate cannot be the dark sector anyway**. Freeing its amplitude to cure the tachyon caps its dark fraction at 2.6×10^{-6} against $\Omega_d = 0.266$ — short by a factor 10^5 . So the object whose background creates the fatal instability is an object that was never going to do the job it was introduced for. Setting $Q_0 = 0$ removes it, and the tachyon gate is then satisfied **identically**.

The price is the whole of §8.1 and §8.2. This is the trade the theory makes, and it should be read as a trade, not as a repair.

5 The kernel, and the repair the theorem forces

5.1 The bounded-boost theorem

The theorem itself is prior work of this programme [23]; what follows uses it and does not re-derive it. Write the acceleration excess as $\Delta \equiv (g_{\text{obs}} - g_{\text{bar}})/a_0$. Because the MOND scalar is *sourced by matter*, Gauss’s law on a sphere gives $J_Y(g_\phi) \cdot g_\phi = g_N$ exactly, with no expansion, so g_ϕ must be a single-valued monotone function of g_N . Writing $g_\phi = a_0 \Delta(s)$, $s = g_N/a_0$, the longitudinal stiffness is exactly $\Sigma_{\parallel} = 1/\Delta'(s)$ — an identity, verified symbolically for arbitrary Δ .

Since Δ is bounded for every kernel of the class, **there is a hard ceiling on the acceleration excess, with no free parameter:**

kernel	ceiling $C = \sup \Delta$
deep-MOND $\sqrt{}$	0.250
standard μ	0.300
exponential carrier, $y e^{-y}$	$1/e = 0.367879$
ν_{RAR} (the operative kernel) [12]	0.647610, at $s = 2.5396$
simple μ	1.000

Λ CDM structurally cannot make this prediction: a halo [13] contributes $g_{\text{obs}} - g_{\text{bar}} = g_{\text{halo}}$, set by M_{200} and concentration, which span decades and are not tied to a_0 . **One system above the ceiling, with a trustworthy baryon model, falsifies the kernel with no fitting freedom.** Currently 99.23% of 2352 SPARC [11] points beyond 2 kpc obey the widest bound at both footings; the eighteen $> 3\sigma$ exceptions lie in five named galaxies, thirteen of them in NGC 5985 alone, and are kept and reported rather than cut.

5.2 The corollary that forces ξ , and the repair that fixes the published kernel

Monotonicity makes $\sup \Delta = \lim \Delta$, so **the Solar-System residual is at least as large as the galactic boost.** Against the binding Solar-System gate — the Pitjev–Pitjeva bound [10] of $6.7 \times 10^{-11} M_{\odot}$ of phantom mass inside Saturn’s orbit, i.e. $4.329 \times 10^{-15} \text{ m s}^{-2}$ — the bare kernel is over by

$$1.4005 \times 10^4 \times (\text{canonical}), \quad 1.6873 \times 10^4 \times (\text{alt}), \quad (6)$$

and never by less than $6.49 \times 10^3 \times$ for *any* kernel in the class. **No kernel of the class can screen the Solar System.** The coherence length ξ is therefore a theorem, not a design choice. At $\xi = 0.10 \text{ pc}$ the operator’s gradient-scale suppression $(\xi/R_{\text{Sat}})^2 = 4.6 \times 10^6$ brings the residual to 3.0×10^{-3} of the bound; at $\xi = 0.15 \text{ pc}$ (alt) to 1.6×10^{-3} .

Two further corrections, both made here:

1. **An attained supremum is inadmissible.** The published kernel is ν_{RAR} up to $s = 2.540$ and then held *flat*. A flat branch has $\Delta' = 0$, hence $\Sigma_{\parallel} = \infty$, hence $J(Y)$ of infinite slope and **no cubic action** at any Solar-System background. The supremum must be **approached and never attained**. Of the five kernels above, only the simple μ is admissible as printed.
2. **The approach rate is capped.** Perturbativity at a planetary source, with ξ at its floor, caps the saturation exponent at $p_{\text{max}} = 1.754$ (both footings; robust to ± 0.35 over two decades

of the $O(1)$ strong-coupling convention, and independent of the radius at which the field is evaluated). Exponential approach is excluded. This constrains the *rate*, never the ceiling — so there is **no** pincer between the bounded boost and the cubic action.

5.3 The kernel the theory actually carries

$$\Delta(s) = C [1 - W(u)^{-p}], \quad u = \sqrt{s}, \quad W(u) = 1 + a_1 u + a_2 u^2, \quad a_1 = \frac{1}{Cp}, \quad (7)$$

with, fitted to the published carried kernel at its own ceiling,

$$C = 0.647610, \quad p = 1.7538, \quad a_2 = 0.9335 \rightarrow \max |\Delta \log_{10} g| = 0.0111 \text{ dex over } s \in [10^{-4}, 10^4]. \quad (8)$$

Properties, each checked:

- **exact deep-MOND coefficient 1** — $Cpa_1 = 1$ by construction, so $\Delta \rightarrow \sqrt{s}$ and $g \rightarrow \sqrt{g_N a_0}$ with no stray 2, $\frac{1}{2}$ or 2π ;
- $\Delta' > 0$ at every finite s (minimum 1.15×10^{-33} over twenty decades), so $\Sigma_{\parallel}$ is finite everywhere and the cubic action exists — the property the published kernel fails;
- **the ceiling is approached, never attained:** at 60 digits, $C - \Delta(10^{14}) = 2.04 \times 10^{-25} > 0$, while $\Delta(10^{14})/C - 1$ is zero to double precision. The bounded-boost prediction and its SPARC test are unchanged;
- $p = 1.7538 \leq 1.754$, at the cap and not beyond it;
- at Saturn's orbit $\Sigma_{\parallel} = 9.4 \times 10^{15}$ (canonical) / 5.6×10^{15} (alt) — **finite**, where the published kernel gives ∞ . Large, and the corresponding longitudinal cone is $\sim 10^{10}c$; that is admissible (the clock's leaves stay spacelike at any *finite* speed, by the identity above) but it is a real oddity of the class and is recorded as one.

The same family fitted to the exponential kernel instead gives $C = 1/e$, $p = 0.844$, $a_2 = 13.31$ and 0.0299 dex.

5.4 A fork that is a measurement, not a convention

Two structures are in play and they are **not** the same theory:

- the **carrier** structure of the action above, where the MOND scalar is sourced by matter, $\Delta(s)$ is monotone, and the boost saturates at $C \cdot a_0$;
- the **AQUAL** structure [2] of the construction's own static limit, where $\mu(y) = 1 - e^{-y}$ acts on the total potential and $\Delta(s) = y e^{-y}$ decreases past $s = 0.63$.

They agree below $s = 0.63$ and diverge above it, because a matter-sourced scalar cannot carry a decreasing Δ . At $s = 3$ the carrier predicts $\Delta = 0.648$ and AQUAL predicts $\Delta = 0.136$ — **0.066 dex in g** , with a maximum separation of **0.073 dex** over $s \in [10^{-4}, 10^4]$, against SPARC's own 0.11–0.13 dex scatter. Detectable in the aggregate, not point by point. And only the carrier arm needs a coherence length at all: the AQUAL arm screens the Solar System by itself.

This fork is unresolved and it is the author's call. It is also a live documentation conflict: the programme's frozen recipe names $\mu(y) = 1 - e^{-y}$ as *the* kernel while the action document carries ν_{RAR} . Until one is amended, any comparison that crosses the two documents is ill-defined.

6 Is the parameter space non-empty? Yes.

Six free parameters $(K_B, c_2, c_{14}, |K_2|, \xi, p)$, with σ derived and gated. Eleven gates, each taken verbatim from its source script. 666,082,872 grid points per footing, plus the $\alpha_2 = 0$ and σ -window loci added explicitly to the c_2 axis so that no narrow band can be stepped over.

gate	admitted fraction (canonical / alt)
H1 PPN α_1	0.387 / 0.387
H2 PPN α_2	0.319 / 0.319
H3 tensor $c_T = 1$ and $G_N > 0$	0.936 / 0.936
H4 BBN	0.691 / 0.691
H5 Solar-System ξ	0.571 / 0.429
H6 saturation $p \leq 1.754$	0.556 / 0.556
H7 clock window $\sigma \in [1.6793, 1.7716]$	0.0120 / 0.0120
H8 Cherenkov (no subluminal mode)	0.410 / 0.410
H9 linear growth	0.402 / 0.402
H10 scalar kinetic health	1 / 1 (sign convention)
H11 clock tachyon	1 / 1 — satisfied identically, because $Q_0 = 0$
INTERSECTION (all eleven)	76,240 / 57,180 points

No subset of the eleven gates up to size four is empty, on either footing. Compare the previous answer, reproduced here as a control: with the condensate present, the intersection was **0 points** on both footings and every minimal incompatible subset contained the clock tachyon.

The explicit point

$$\begin{aligned}
K_B &= 0.2 \\
c_{14} &= 1.0000 \times 10^{-6} \\
\sigma &= \sigma_* = 1.679312732 && \text{(a design choice inside the window, at its lower edge)} \\
c_2 &= 1.679318 \times 10^{-6} && = 2\sigma c_{14} / (2 - c_{14} - 3\sigma c_{14}), \text{ determined by } \sigma \\
|K_2| &= 1.9294 \times 10^6 && = (2 - K_B)^2 / c_2, \text{ exactly on the closure locus} \\
\xi &= 0.10 \text{ pc (canonical)} / 0.15 \text{ pc (alt)} && \text{— at the theorem-forced floor} \\
p &= 1.7538 && \text{— at the cap} \\
Q_0 &= 0 && \text{— no condensate}
\end{aligned}$$

Derived:

$$\begin{aligned}
\alpha_1 &= -4.48 \times 10^{-6} \text{ (canonical)} / -4.25 \times 10^{-6} \text{ (alt)} && \text{bound } 10^{-4} \\
\alpha_2 &= -2.02 \times 10^{-7} \text{ (both)} && \text{bound } 4 \times 10^{-7} \text{ (margin } 1.98\times) \\
\alpha_3 &= 0, \quad \gamma = 1, \quad c_T = c \text{ exactly} \\
c_{s,\text{mix}}^2 &= 1.679318, \quad S_{\text{eff}} = 0.000000 \\
G_{\text{cos}}/G_N &= 0.999997481, \quad G_N/G = 1.000000500
\end{aligned}$$

This point is **derived**, not read off the grid, and then confirmed admitted by all eleven gates on both footings.

Two structural results that fell out of the assembly

(a) **The “closure locus” has a meaning.** The relation $c_2|K_2| = (2 - K_B)^2$, previously reported as a tuning that must hold to between 4% and 38%, is *exactly* the locus on which the mixed clock–scalar mode travels at the clock’s own speed: $S_{\text{eff}} = 1 - c_{s,\text{mix}}^2/\sigma$ to leading order in c_{14} and c_2 (verified to 3×10^{-6} relative at the exhibited point). It is still a relation between two otherwise-free parameters and is still reported as a tuning — but it is a *speed-matching* condition, not an arbitrary one.

(b) **A derived ceiling on the theory’s own cosmology.** Cherenkov safety ($c_{s,\text{mix}}^2 \geq 1$) together with the clock window ($\sigma < 1.7716$) cap the surviving linear cosmological MOND source at

$$S_{\text{eff}} \leq 1 - \frac{1}{\sigma} = 0.4355, \quad (9)$$

so **the action can never deliver more than 44% of its own naive linear source**, whatever the parameters. This was an input to nothing; it is a consequence.

Robustness

Two of the eleven gates are conditional. Dropping either only enlarges the region:

gate set	canonical	alt
all eleven	76,240	57,180
without H7 (the σ identification, conditional on the unpublished variable map)	7,567,560	5,675,670
without H8 (Cherenkov, which may not apply at all to a conformally coupled clock)	309,400	232,050

The non-emptiness does not rest on either conditional.

7 The gate table

Verdicts at the exhibited point, both footings. Full table with sources in `L43_assemble_theory.out`, Part V. Tally over both footings: **42 PASS, 18 FAIL, 8 other** (pending / diagnostic / not a discriminant).

gate	canonical	alt	number
Cassini quadrupole	PASS	PASS	ξ is $3.3\times / 3.0\times$ the exact fourth-order solve’s own floor
Saturn phantom mass	PASS	PASS	bare kernel $1.40 \times 10^4 \times / 1.69 \times 10^4 \times$ over; screened to $3.0 \times 10^{-3} \times / 1.6 \times 10^{-3} \times$
Sunward anomaly	PASS	PASS	$1.3 \times 10^{-17} / 7.0 \times 10^{-18}$ vs bound $3.66 \times 10^{-14} \text{ m s}^{-2}$
PPN α_1	PASS	PASS	$-4.48 \times 10^{-6} / -4.25 \times 10^{-6}$ vs 10^{-4}
PPN α_2	PASS	PASS	-2.02×10^{-7} vs 4×10^{-7} (margin $1.98\times$)
PPN α_3	PASS	PASS	0 exactly

gate	canonical	alt	number
PPN γ	PASS	PASS	1 exactly
Tensor speed (GW170817)	PASS	PASS	$c_{13} \equiv 0 \Rightarrow c_T = c$ exactly at every K_B
Newton constant positivity	PASS	PASS	$G_N/G = 1.0000005$
DOF count	FAIL vs “$N_{\text{grav}} = 2$” as a total count	same	4 = 2 tensor + 1 clock + 1 scalar. Passes the requirement as written (separately counted and healthy); fails it read as a total
DOF health	PASS	PASS	clock $A_0 = 0.4615 > 0$, positive mode energy, hyperbolic at σ_* ; scalar Bogoliubov $\omega^2 = c_s^2 k^2 (1 + \xi^2 k^2) > 0$ $S_4 > 0$ at 59/59 branch points at σ_* ; growth $\rightarrow$ bounded oscillation
Hadamard well-posedness	PASS	PASS	MOND at the action’s own a_0 , coefficient exactly 1.000000
Galactic rotation curves	PASS	PASS	$\Phi = \Psi$ to $< 10^{-4}$ out to 1 Mpc
Lensing/dynamics slip (galaxies)	PASS	PASS	
Bounded-boost ceiling on SPARC	PASS	PASS	99.23% of 2352 points; 18 exceptions in 5 named galaxies
Kernel tightness vs a fitted halo	NOT A DISCRIMI- NANT	same	0.142 vs 0.171 dex at zero parameters, but a prior-shrunk NFW reaches 0.085
Gaia DR4 wide binaries	PENDING	PENDING	γ_v ceiling 1.0450 / 1.0300, registered; Arm A band 1.16–1.23
Clusters: mass at R_{500}	FAIL	FAIL	required/delivered 1.618 ± 0.022 / 1.493 ± 0.020 (dynamics); 1.987 ± 0.246 / 1.834 ± 0.227 (lensing)
Clusters: bounded-boost ceiling	FAIL	FAIL	excess $3.37 a_0$ at 40 kpc = $5.2\times$ the ceiling
Clusters: lensing SHAPE	FAIL	FAIL	$\Delta\Sigma$ short 2.5–2.7 $\times$; log-slope $+0.53 \pm 0.06$ shallower = 9σ
Cluster residual = cosmic share	DIAGNOSTIC	DIAGNOSTIC	required 5.73 ± 0.68 vs $\Omega_{\text{dm}}/\Omega_b = 5.43$, universal to 12%
Coma ultra-diffuse galaxies	FAIL	FAIL	+1.196 dex (factor 14) at 4.9σ against the coherent systematic floor
Binary galaxies (2MRS)	FAIL alone	FAIL alone	isolated deep-MOND amplitude 1.802 ± 0.041 ; 1.141 ± 0.028 with a cosmic share. Significance is statistics-only: 19.6σ falls to 7.2σ with a 0.10 dex mass-to-light floor and 3.0σ with the isolation band. Direction survives, number falls.
BBN	PASS	PASS	$K_B = 0.2 \leq 0.25$; $G_{\text{cos}}/G_N - 1 = -2.5 \times 10^{-6}$ vs 0.13
CMB acoustic peaks	FAIL	FAIL	no dark component in the action once $Q_0 = 0$
Linear growth	PASS as an equation	same	$S_{\text{eff}} = 0$ exactly on the closure locus — and nothing to grow
σ_8 / S_8	FAIL (for the late-roll route)	FAIL	$S_{8,\text{eff}} = 1.06\text{--}1.20$, $+9\sigma$ to $+34\sigma$

gate	canonical	alt	number
Gravitational Cherenkov	PASS	PASS	clock 29.6% superluminal; the bound constrains slow modes
Causality / no CTC	PASS	PASS	$G^{\mu\nu}n_\mu n_\nu = -1/\sigma < 0$ for every $\sigma > 0$
Black holes	PASS	PASS	universal horizon $r = 3M/2$; even an infinite-speed mode trapped
Longitudinal cone at Saturn	PASS (repaired)	PASS (repaired)	$\Sigma_{\parallel} = 9.4 \times 10^{15} / 5.6 \times 10^{15}$ — finite, where the published kernel gives ∞
Strong coupling at a planet	PASS at $p \leq 1.754$	same	$\delta g/g_* = 1520$ for Earth without ξ ; perturbative with ξ at its floor
a_0 - Λ tie	FITTED	FITTED	$\kappa = 0.4998 / 0.6023$
a_0 universality	FAIL	FAIL	the ladder spans 0.78 dex across system classes

Read the table this way: **every row that lives inside a galaxy or the Solar System passes; every row above a galaxy fails**, and the failures are not independent.

8 The holes, with sizes

8.1 H-A The cluster deficit — the big one

The residual behaves like **mass** in dynamics *and* in lensing, so no lensing-sector repair can reach it. Five X-COP clusters [14, 15] with published weak lensing [31], at each cluster’s own R_{500} : required/delivered against hydrostatic mass 1.618 ± 0.022 (**canonical**) / 1.493 ± 0.020 (**alt**); against weak lensing 1.987 ± 0.246 / 1.834 ± 0.227 ; and the two probes agree with each other ($S_{\text{lens}} - S_{\text{dyn}} = +0.37 \pm 0.24, 1.55\sigma$). **The shortfall’s own 13–15 σ is statistics-only: a 10% coherent X-ray mass calibration moves it 22%, giving 4.4 σ . The shortfall itself is unaffected.** Four independent lensing teams give 1.623 ± 0.080 / $1.499 \pm 0.074, 1.4\sigma$ away. A cored lensing-derived residual has been reported for MOND-class kernels [32]; §?? finds that disagreement to be a parametrisation artefact rather than a physical conflict.

The information dynamics could not give is **shape**. In raw $\Delta\Sigma$ over 0.5–2 Mpc the framework is short by 2.5–2.7 $\times$ and its $\Delta\Sigma$ log-slope is $+0.53 \pm 0.06$ **shallower** — a 9 σ **error** — because its phantom is a near-uniform sheet. Machinery validated by pushing the *measured* X-ray profile through it, which reproduces measured $\Delta\Sigma$ to a median 1.11; analytic NFW $\Sigma/\Delta\Sigma$ to 1.3×10^{-6} . The NFW-refit shape systematic is 0.43, so the quoted shortfall **understates** the disagreement.

And the kernel’s own ceiling is violated: the excess is 3.37 a_0 at 40 kpc, 5.2 $\times$ **the operative ceiling**, so no interpolation function can absorb it. Both escapes are closed quantitatively — non-thermal support would need $\sigma_{1D} = 673\text{--}1155 \text{ km s}^{-1}$ against Hitomi/XRISM’s $164 \pm 10 \text{ km s}^{-1}$ in Perseus [16]; unseen baryons would need 3.9–5.2 $\times$ the imaged X-ray gas.

Every mechanism the programme owns has been closed against it, one at a time: an interpolation kernel (clusters need 2.2–5.1 $\times$ the galaxy boost at the same acceleration, $|z| = 13$); a fixed-strength finite-range force (BBN, 41 $\times$); a screened force in Φ , ρ or M (100% density overlap at 12.8 σ , plus a cosmological ordering argument); hydrostatic bias (the required b is **negative**, -0.82 , against a measured $[0, +0.42]$; a negative b means $\sigma^2 < 0$ in 12 of 12 clusters); the solenoidal field (real, 14–27% locally inside a merging pair, but $\text{div } a_S = 0$ forces $\langle a_S \cdot \hat{r} \rangle \equiv 0$ at every radius — it closes 3.9% of the gap and runs the wrong way); a late-time roll of the coupling (gives 0.97 of the

required 2.2–5.1 contrast, and cancels exactly from the growth source in a consistent scalar–tensor model); light thermal relics (Tremaine–Gunn floor [17] $m \geq 4.67$ eV, and an N_{eff} -vs-RAR pincer with no interior at 27.6 vs 11 eV); wave dark matter (pincer with no interior, factor 1.4 in mass); four separate condensate constructions; an environment switch.

And the required ratio is not monotone in scale, which a scale-free kernel forbids: 1900 isolated 2MRS [18] major pairs need $M_{\text{dark}}/M_{\text{bar}} = 30.9 \pm 1.6$ within the pair separation against 5.73 ± 0.68 at $0.80 R_{500} - 5.7\times$. **The 16.3σ once quoted is against the cosmic 5.43, not against 5.73 ± 0.68 (which gives 14.5σ), and it is statistics-only on 1900 pairs sharing one mass-to-light scale, one distance scale and one isolation criterion; with a coherent floor it is 3.1σ .** The deficit *grows* with mass at 3.9σ , itself statistics-only.

What satisfies all four cluster requirements at once — the mass, the phase-space density, near-absence in galaxies, and a flat-or-falling enclosed ratio — is a cold, baryon-tracing component. The theory does not contain one. That is recorded as a real cost to the modified-gravity programme, not as an open question.

8.2 H-B No dark sector at all, hence no CMB and no structure

Removing the condensate is what cures the clock tachyon, and it was the action’s only dark component. Size: the condensate’s dark fraction caps at 2.6×10^{-6} against $\Omega_d = 0.266$ — short by $10^5\times$. With it gone the theory has baryons only; the CMB third peak [19], S_8 , and the matter power spectrum are unaddressed. The MOND scalar’s own causal growth boost does **not** regenerate $P(k)$ at linear order: $\sigma_8 \leq 0.65$ with a 20–2000 $\times$ deficit at $k = 0.5\text{--}1 h \text{Mpc}^{-1}$, at any $|K_2|$, on both footings. The one untested route is nonlinear top-down fragmentation, which a per-mode linear calculation cannot represent.

8.3 H-C $\kappa = \frac{1}{2}$ is fitted

$a_0 = \kappa c \sqrt{G\rho_\Lambda}$ is dimensionally forced; κ is not. Recomputed here from Planck’s Ω_Λ [19]: $c\sqrt{G\rho_\Lambda} = 1.87251 \times 10^{-10} \text{ m s}^{-2}$, so $\kappa = 0.4998$ (**canonical**) and 0.6023 (**alt**) — the tie does not even pick a footing.

Six theorems constrain any derivation. The decisive one: in **any** local action whose MOND function multiplies $\sqrt{-g}$ with a field-independent coefficient, $F \rightarrow F + C$ adds exactly $C\sqrt{-g}$ and is exactly degenerate with Λ . **κ is a zero mode of the whole class** — verified on four structurally different Lagrangians, independent of the function, the field count and the coefficient. No curvature polynomial takes a half-integer power of Λ on de Sitter, so $a_0 \propto \sqrt{\Lambda}$ *requires* an order parameter of odd mass dimension whose square is the vacuum energy; any 4-volume average of a gradient sector is driven to zero in a Λ -dominated future (10^{-8} by $10 t_0$), closing the global-constraint class; boundary and Brown–York terms see only $J(0)$, which the zero-mode theorem already makes pure Λ . Deriving κ reduces to fixing one number, $\beta/\sqrt{\tilde{Z}} = \kappa/\sqrt{2} = 0.354$. **A binding guard: 27 simple numbers lie inside the measured 3σ band. Landing in the band is not evidence — only a derivation is.** And κ is not the discriminating observable: total comparison uncertainty is 8.0% and nothing in $[0.40, 0.66]$ is rejectable at 3σ .

8.4 H-D The foliation’s escape is undecided

See §2. If temporal nonlocality survives a Hamiltonian mode count, the preferred foliation is a choice rather than a theorem. Being decided in a parallel effort; **not** claimed here either way.

8.5 H-E The variable map is unpublished

c_{14} has three inconsistent readings in the construction’s files spanning 0.073–1.333, and c_2 vanishes at the state where the mode was measured. The $\sigma \leftrightarrow (c_{14}, c_2)$ identification of §4.1 is therefore stated, not proved. The region is non-empty with or without it; the sharpness of $c_{14} \leq 2.0 \times 10^{-6}$ is what depends on it.

8.6 H-F c_2 sits five orders below the certified healthy range

The scalar sector’s linear health was certified for $c_2 \in [0.01, 0.1]$. The exhibited point has $c_2 = 1.7 \times 10^{-6}$. Five orders of untested parameter. Not a computed failure — a gate nobody has run.

8.7 H-G Two documentation conflicts are open

(i) the kernel: the frozen recipe names $\mu = 1 - e^{-y}$, the action document carries ν_{RAR} ; they differ by up to 0.073 dex and the exponential ceiling is exceeded $\sim 5\times$ more often on the bulgeless SPARC control. (ii) ξ : the action document’s parameter table still quotes 0.03/0.05 pc — the *unsaturated* partner’s floors — against 0.10/0.15 pc for the kernel a scalar can actually carry. One document each needs amending.

8.8 H-H Standing liabilities the assembly does not repair

Coma ultra-diffuse galaxies [29]: the amplitude stands at $+1.196 \pm 0.062$ dex (a factor of 14), but the significance is 4.9σ , **not the 19.4σ once quoted** — the 0.062 dex bar is statistics-only on eleven objects sharing one M/L scale, one estimator and one cluster model, and the coherent systematic floor is 0.227 dex. Closing it needs $M/L \times 14.4$, and tides are excluded by a dedicated MOND simulation [30]. The a_0 ladder spans 0.78 dex across system classes. Unequal-mass wide binaries carry an open numerical item. Non-spherical accretion and the existence theory of the non-spherical fourth-order law are untouched.

9 What is distinctive, with numbers, and what would falsify it

1. **The bounded boost.** $g_{\text{obs}} - g_{\text{bar}} \leq C a_0 = 6.06 \times 10^{-11} / 7.30 \times 10^{-11} \text{ m s}^{-2}$, *everywhere, in every system, with no free parameter.* ΛCDM cannot make this prediction. **Falsified by one system above the ceiling with a trustworthy baryon model.**
2. $a_0(z)$ — **the surviving distinctive prediction.** The deep-MOND baryonic Tully–Fisher zero point is flat to $< 1\%$ out to $z = 5$ on the derived law, against ΛCDM ’s $+0.33$ dex by $z = 2.5$. **Decisive measurement: a deep-MOND lensed rotator at $z \approx 2$ – 2.5 to ± 0.13 dex,** pre-registered in [26]. This, not κ , is the discriminator.

3. **Gaia DR4 wide binaries — and it is a Newton-like answer.** $\gamma_v \leq 1.0450$ (canonical, $\xi = 0.10$ pc) / **1.0300** (alt, $\xi = 0.15$ pc), falling toward 1 as ξ grows, against a separately registered band of 1.16–1.23 [20, 21] and Newton’s 1.000. These are **ceilings**: DR4 can kill this arm from above, it cannot confirm it over Newton. **A measurement inside 1.16–1.23 falsifies the coherence-length structure.**
4. **The saddle null.** No Bekenstein–Magueijo anomaly [22] at any Solar-System field null — the coherence operator erases it by 10^{-29} . A LISA-Pathfinder-class saddle flyby detecting one falsifies the operator.
5. **The saturation exponent is bounded above.** $p \leq 1.754$: the approach to the ceiling is a power law, never exponential. Measuring a faster approach falsifies the perturbative existence of the theory’s own cubic action.
6. **The carrier/AQUAL fork, priced here for the first time.** 0.073 dex maximum separation, 0.066 dex at $s = 3$. BIG-SPARC’s transition bins decide which structure the data prefer — and only the carrier arm needs a coherence length at all.
7. **A prediction the removed branch would have made, listed to be honest about what removing it costs.** If any component with the scalar’s stiffness supplied part of the cluster residual, its peak radius would be set by $H = 0.42 e c^2 / (|K_2| a_0)$ alone and would therefore be **the same physical radius in every cluster**, where an NFW scale radius grows as $M^{1/3}$. The assembled theory has no such component.

10 Verdict, in three sentences

The admissible region is non-empty, and at an explicit point inside it this is a complete, internally consistent, Solar-System-safe, galaxy-correct relativistic theory of gravity — one metric, matter minimally coupled, exactly Einstein in the metric sector, $c_T = c$ exactly, four healthy propagating modes, a preferred foliation forced by a theorem rather than assumed, and Milgrom’s law at the framework’s own a_0 with coefficient exactly 1 and no lensing/dynamics slip.

It is **not** a complete theory of the universe: curing its one fatal instability required deleting its only dark component, so it is short by a factor of 1.49–1.99 in mass at R_{500} in both lensing and dynamics, wrong by 9σ in the shape of $\Delta\Sigma$, over its own hard acceleration ceiling by $5.2\times$ in cluster cores, and silent on the CMB, S_8 and the matter power spectrum.

What is missing is exactly three things and they should be named that plainly: a cold, baryon-tracing component that every mechanism this action contains has been shown one by one unable to supply; a derivation of κ , which a zero-mode theorem proves no local action of this class can give; and a decision on the locality hypothesis that the foliation theorem rests on.

Reproduction and reproducibility

The assembly is one committed script, run as

```
python3 fable_independent_2026/L43_assemble_theory.py
```

Exit 0. Nine controls (K1–K9) rebuild six settled numbers from their own algebra and confront them with the value printed in the source lane’s own `.out`; two more reproduce a known PASS and a known FAIL from existing gate scripts. Seven kernel checks (N1–N7), thirteen gate-fraction checks, and the admissibility test (T1, T2) follow. Every control passes. The four FAIL lines — T5, T7, T8, V1 — are results, not machine errors: the documentation conflicts, the failures above a galaxy, the cluster deficit, and the incompleteness itself.

Supporting lanes, each with its own script and `.out` in the same directory of the public repository: L4, L8 (verification of the construction), L11 (the MOND limit), L13 ($\alpha_1 = -4c_{14}$), L14 (the previous, empty sweep), L19 (Cherenkov applicability), L21 (binary galaxies), L22 (the curl field), L23 (Coma UDGs), L24 (lensing vs dynamics), L26 ($\sigma > 1$), L27/L31 (the foliation theorems), L28 (tightness), L29 (S_8), L32 (κ), L33 (the scalar cone), L34 (the bounded-boost audit). The adversarial record audit is `L36_record_audit.py` (commit `fc47e26ca`); the assembly is commit `94c41856e` and the significance corrections carried into this paper are commit `e8ae735f1`. Both a_0 footings (9.3619×10^{-11} and $1.1279 \times 10^{-10} \text{ m s}^{-2}$) are carried throughout. Every number is produced by a committed script with checks that can fail. This is a preprint of an AI-assisted draft; it is not peer reviewed. It claims no derivation of a_0 or of $\kappa = \frac{1}{2}$, no dark sector, and no empirical detection.

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